

Neuro-Symbolic Music Recommendation with a Knowledge-Graph-Informed Heterogeneous Graph Transformer

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Abstract

We present a deep-learning pipeline for content- and preference-aware music recommendation built on a heterogeneous music knowledge graph (KG). Starting from the Lakh MIDI and Million Song datasets, we align 7,013 tracks, compress their 1,526-dimensional symbolic feature vectors to a 128-dimensional embedding with a fully connected autoencoder (final MSE 0.616), and pre-train KG embeddings over 807,955 relational triples — RotatE (with ComplEx as a reference) — using RotatE to initialise every entity type. A Heterogeneous Graph Transformer (HGT) is then trained with an intensity-weighted listwise loss augmented by logit adjustment for popularity debiasing. Against three baselines (MostPopular, KNN collaborative filtering, and an XGBoost content-behaviour hybrid), the HGT reaches Recall@10 0.105, NDCG@10 0.097 and 79.7% catalogue coverage with the lowest popularity bias of all models (0.053), giving an Overall Score of 0.399 — second only to KNN-CF (0.421) and significantly ahead of the XGBoost hybrid (0.336) and the popularity baseline (0.148). Relative to our Milestone 2 report, evaluating the HGT with message passing consistent with training (undirected) resolves the directed-evaluation artefact that had previously collapsed its scores, and a retuned architecture plus RotatE initialisation now outperform the earlier configuration on the composite objective. We further analyse the learned latent space (20 emergent user archetypes) and produce per-edge attention rationales that explain individual recommendations.

1. Introduction

Music recommendation is dominated by collaborative filtering (CF), which infers taste from interaction overlap alone. CF favours popular content and is blind to *why* a user likes a track: a listener of Baroque counterpoint and a listener of jazz both interact with mainstream pop, so a purely behavioural model conflates two distinct preferences. We ask whether a deep model that reasons over explicit musical structure — tracks, artists, genres, keys, tempo classes, in-

struments, and users connected by typed relations — can recommend more diverse and content-grounded items while remaining competitive on ranking quality.

Plan. Our approach is an end-to-end deep-learning pipeline (Fig. 1). We (i) integrate the Lakh MIDI Dataset [1] and Million Song Dataset [2] into a music KG; (ii) compress high-dimensional symbolic descriptors with a *symbolic-feature autoencoder* and pre-train KG embeddings (RotatE [3], with ComplEx [4] as a reference) to initialise all node types; (iii) train a *Heterogeneous Graph Transformer* [5] with a popularity-debiased listwise loss; and (iv) evaluate the model family against three baselines on a ranking-plus-diversity metric suite, complemented by latent-space and attention-based interpretability analyses.

This document is the final Deep Learning report. It consolidates the proposal and the two milestone reports and updates them to match the work actually done. The single most consequential change since Milestone 2 is that the HGT — previously crippled by an evaluation-protocol mismatch — now ranks second among all models, with the lowest popularity bias and strong catalogue coverage, once inference is made consistent with training and the architecture is retuned. Our contributions are: (1) a deep representation stack coupling a symbolic autoencoder, transductive KG embeddings, and a heterogeneous GNN; (2) an intensity-weighted listwise loss with logit adjustment that exposes a tunable ranking-versus-diversity trade-off; (3) a controlled study of node initialisation, debiasing strength, and architecture; and (4) interpretable recommendation rationales grounded in HGT attention.

2. Problem Statement

Task. Given a user’s listening history, recommend a ranked list of K unseen tracks. We treat this as implicit-feedback top- K ranking over a fixed catalogue of 6,054 modelled songs and 284,864 users.

Data. We use the Lakh MIDI Dataset (symbolic MIDI aligned to audio via chroma DTW) and the Million Song Dataset (audio-derived scalars, artist/genre metadata, and

the Echo Nest Taste Profile of $\sim 48\text{M}$ user–song–playcount triplets). After mismatch removal, an inner join to the LMD-matched tracks, and a 5-interaction cold-start floor, 3,048,959 interactions remain over 285,037 users and 7,227 songs; 7,013 tracks yield valid symbolic features and form the modelling set (Sec. 4).

Expected result. A model that exploits musical content and relational structure should (a) match or approach CF ranking quality while (b) substantially improving catalogue coverage and lowering popularity bias, and (c) offering human-readable explanations — properties that popularity-driven CF cannot provide.

Evaluation. A user-level stratified 70/10/20 split (by primary genre and release decade) preserves popularity and content distributions across folds; KG structure and track features are visible to all folds and only interactions are partitioned. We report per-user Recall, Precision, F1, HitRate, MRR, and NDCG at $K \in \{5, 10, 20, 50, 100\}$, plus two catalogue-level metrics — coverage $\text{Cov} = |\bigcup_u R_u|/|I|$ and popularity bias $\text{PB}@K$ — fused into a composite

$$\text{Overall}@K = 0.6 \text{NDCG}@K + 0.2 \text{Cov} + 0.2(1 - \text{PB}@K), \quad (1)$$

which rewards accurate *and* diverse recommendation and serves as the HGT checkpoint criterion. Significance is assessed with paired Wilcoxon signed-rank, t -tests, and bootstrap confidence intervals over a shared 284,864-user evaluation set.

3. Related Work

Collaborative and neural CF. User–user/item–item similarity and matrix factorisation, often optimised with BPR [6], dominate implicit feedback; Neural CF [7] replaces the inner product with an MLP. These methods are content-agnostic and amplify popularity bias — the gap our model targets. We use KNN-CF and a popularity baseline as references.

Content-based and hybrid. jSymbolic [8] and music21 [9] extract hundreds of symbolic descriptors from MIDI; their high dimensionality and redundancy motivate autoencoder compression before recommendation. Our XGBoost hybrid retrieves on autoencoder embeddings and re-ranks with a gradient-boosted LTR model, isolating the value of graph structure.

KG-enhanced recommendation. KGCN [10] and KGAT [11] propagate KG semantics into item representations; LightGCN [12] shows that minimal propagation suffices for homogeneous user–item graphs; Liu et al. [13] combine KG features with multi-task learning for music. We differ by building the KG from scratch, by using KG embeddings as *node initialisers* for a heterogeneous GNN

Table 1. Interaction filtering pipeline.

Step	Interactions	Users	Songs
Raw	48,373,586	1,019,318	384,546
–Mismatches	45,795,100	1,019,318	378,309
$\cap$ LMD	4,243,384	848,243	7,465
–Cold-start	3,048,959	285,037	7,227

rather than as a separate aggregation module, and by injecting 1,526-dimensional symbolic content the KG structure cannot capture.

Heterogeneous GNNs and KG embeddings. The Heterogeneous Graph Transformer [5] learns per-(source,edge,target)-type transformation and attention, preserving typed structure and exposing inspectable attention. Complex-valued KG embeddings RotatE [3] and ComplEx [4] model asymmetric and compositional relations; we train them on the non-interaction triples and use them to initialise every entity type, including structural nodes (genres, keys, instruments) that have no other numeric representation.

4. Data Pipeline and Preprocessing

Alignment. MIDI files are matched to MSD tracks using precomputed chroma DTW similarities [1]; we keep matches with score ≥ 0.55 and deduplicate, yielding $\sim 19\text{k}$ unique tracks.

Interaction filtering. Raw Taste Profile interactions (48,373,586) are cleaned in three stages (Tab. 1): MSD mismatch IDs are removed, interactions are inner-joined to the LMD-matched track set, and users with fewer than five remaining interactions are dropped to mitigate cold-start, leaving 3,048,959 interactions over 285,037 users and 7,227 songs.

Symbolic features. jSymbolic and music21 are applied to all MIDI files; 7,013 tracks succeed, each described by 1,526 numerical descriptors spanning pitch, melody, harmony, rhythm, dynamics, texture, and instrumentation. Features are standardised before autoencoding.

Knowledge graph. Tracks, performances, artists, genres, instruments, keys, modes, tempo classes, and decades become typed nodes; user listening events become $\text{user} \rightarrow \text{track}$ edges. The populated graph contains 1.9×10^7 triples; the entity composition is 285,037 listeners, 7,013 tracks/performances, 3,668 artists, 645 genres, 129 instruments, 12 keys, 9 tempo classes, 7 decades. The KG is loaded into GraphDB for validation and streamed into a PyTorch Geometric HeteroData object for learning. (KG

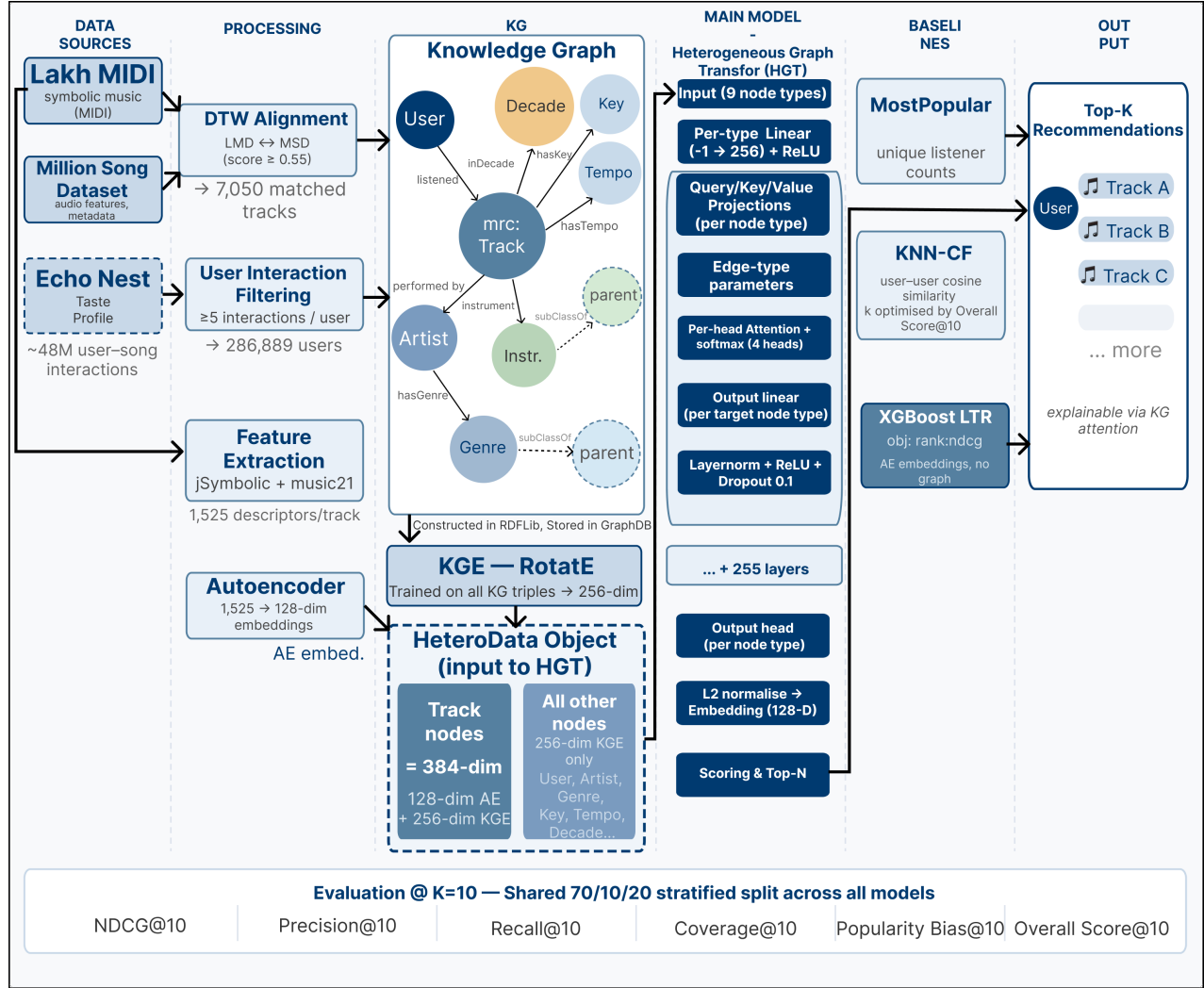


Figure 1. End-to-end methodology. Three data sources (Lakh MIDI, Million Song, Echo Nest Taste Profile) are aligned and filtered; 1,525 symbolic descriptors per track are compressed to 128-D by an autoencoder; an RDF knowledge graph is built in GraphDB; RotatE KG embeddings (256-D) initialise all node types; the HGT consumes a 9-node-type PyTorch Geometric HeteroData object (track nodes 384-D = 128-D AE || 256-D KGE) and is trained against three baselines; all models share a 70/10/20 stratified split and are evaluated at $K=10$.

design and SPARQL validation are detailed in the companion Knowledge Graphs report; here we focus on the learning stack.)

KG statistics. The fully populated graph spans 1.91×10^7 triples; `rdf:type` (612,827) and `instrument` (56,891) dominate the relation distribution, followed by `genre`, `performance`, and `listening` links (Tab. 6, Appendix). Genre tags are heavy-tailed — led by *rock* (2,117 artists), *pop* (1,062), and *electronic* (1,037) — and high-confidence key detections recover the Western tonal prevalence of C (211), G (196), and D (150). The most-reached track is *Secrets* (OneRepublic, 50,058 distinct listeners), while the highest total play

count is *You're The One* (Dwight Yoakam, 390,283 plays). The schema-level ontology (TBox) and a concrete track neighbourhood (ABox) are shown in Figs. 9–10 (Appendix).

Splits. A single user-level stratified split (by primary genre and decade, 2,186 strata) gives 1,943,681 train, 348,054 validation, and 559,806 test interactions over 284,864 users and 6,054 evaluated songs.

5. Models and Technical Approach

5.1. Symbolic-Feature Autoencoder

A fully connected autoencoder compresses the standardised 1,526-dimensional symbolic vector to a 128-dimensional bottleneck. The encoder is $1526 \rightarrow 512 \rightarrow 256 \rightarrow 128$ with a symmetric decoder, each hidden layer followed by ReLU and dropout 0.2. The bottleneck embedding serves both as the track node feature in the GNN and as the item/user representation in the hybrid baseline. It minimises reconstruction MSE.

5.2. Knowledge-Graph Embeddings

Our KG is dominated by *asymmetric* relations (a user listens to a track, not the reverse), which real-valued translation models cannot represent. **RotatE** encodes each relation as a complex rotation ($h \circ r \approx t$), naturally capturing asymmetry, inversion, and *compositional* chains (e.g. a decade predecessor/successor traversal); it is our primary initialiser. **ComplEx** achieves asymmetry by conjugation and is retained for comparison. Both are trained on 807,955 triples (listening edges excluded to prevent target leakage), entity_dim 128, producing 256-dimensional vectors for all 304,110 entities.

5.3. From KG to Learnable Tensors

Each ontology class becomes a node type and each predicate a typed directed edge. Track nodes are initialised with the 384-dimensional concatenation of the 128-dim autoencoder embedding and the 256-dim RotatE vector; all other node types receive their 256-dim KG embedding. Because listening events are direct `user`→`track` edges, user-to-item message passing requires a single hop. To let the user node receive and emit messages symmetrically, message passing is performed on the *undirected* graph; this consistency between training and inference is the fix for the Milestone 2 directed-evaluation artefact.

5.4. Heterogeneous Graph Transformer

The HGT [5] stack uses 2 message-passing layers, 4 attention heads, hidden dimension 256, and dropout 0.1, learning separate transformation and attention parameters per (source-type, edge-type, target-type) triple (see the main-model column of Fig. 1). Each HGTConv layer applies per-node-type Q/K/V projections and per-edge-type message/attention weights, computes 4-head attention over typed neighbours, aggregates, and passes through a per-target-type output linear, LayerNorm, ReLU, dropout, and a residual connection. A per-type MLP head ($256 \rightarrow 256 \rightarrow 128$) projects the final hidden state to a 128-dimensional output that is ℓ_2 -normalised; scoring is cosine similarity between user and track embeddings with seen items masked.

Loss. For normalised user embedding $\hat{\mathbf{u}}$ and track embeddings $\hat{\mathbf{t}}_i$, the logit-adjusted score is

$$s_i = \frac{\hat{\mathbf{u}}^\top \hat{\mathbf{t}}_i}{\tau} + \lambda \log p_i, \quad (2)$$

where $\tau=0.1$ sharpens the softmax and p_i is the Laplace-smoothed marginal listen probability. Since $\log p_i < 0$, popular tracks are penalised and long-tail items are surfaced. With soft targets $\tilde{p}_{ui} \propto \log(1 + c_{ui})$ (play-count intensity, row-normalised), the objective is the listwise cross-entropy

$$\mathcal{L} = -\frac{1}{|B|} \sum_{u \in B} \sum_i \tilde{p}_{ui} \log \text{softmax}(\mathbf{s}_u)_i. \quad (3)$$

Checkpoints are selected by Overall@10 (Eq. 1) on validation.

5.5. Baselines

MostPopular ranks unseen items by global play frequency, identically for all users. **KNN-CF** is a user–user collaborative filter on the ℓ_2 -normalised interaction matrix; $k=6$ is selected by a validation sweep. **XGBoost Hybrid** retrieves 200 candidates per user via autoencoder-profile cosine similarity and re-ranks with a gradient-boosted LTR model (`rank:ndcg@10`) on 300-dimensional feature vectors (128-dim user-profile || 128-dim track AE || 44 categorical features: one-hot key/mode/tempo class, target-encoded genre/artist, multi-hot instruments), fusing content and behaviour without graph structure.

6. Experiments

We run the following experiments, all on the splits of Sec. 4:

- **Autoencoder compression:** 200-epoch transductive run over 6,054 tracks; per-epoch MSE monitored.
- **KGE training:** RotatE and ComplEx trained (200 epochs each) and compared at the embedding-loss level; RotatE alone initialises the HGT (ComplEx is a reference only).
- **Architecture search:** HGT sweeps over width, heads, layers, dropout, learning rate, weight decay, and temperature, ranked by Overall@10 on validation.
- **Initialisation ablation:** RotatE vs. random node features at the selected configuration.
- **Debiasing (λ) ablation:** $\lambda \in \{0, 0.1, 0.2, 0.3, 0.5\}$ to characterise the ranking-versus-coverage trade-off.
- **Multi- K benchmark:** all four models at $K \in \{5, 10, 20, 50, 100\}$ with paired significance testing.
- **Interpretability:** latent-space clustering (GMM over UMAP) and per-edge attention rationales.

Training budgets and hardware are summarised in Tab. 2.

Table 2. Model and training budget. Input shape is per-example. AE / KGE / XGBoost wall-clock is logged; the HGT was trained for 300 epochs (best checkpoint at epoch 240) on the same A100 class but per-run wall-clock was not separately logged. Hardware: NVIDIA A100 80 GB (A2 Ultra) for most runs; NVIDIA RTX PRO 6000 (G4) for a subset.

Model	Architecture / depth	Input	Loss	Optimiser	Epochs	LR	Batch	Time
Autoencoder	MLP 1526-512-256-128 (+sym. dec.)	1526	MSE	Adam	200	10^{-3}	2048	≈ 11 s
RotatE (KGE)	complex rotation, dim 128 ($\rightarrow 256$)	triple	neg.-sampling rank	Adam	200	10^{-2}	32,768	44 m
ComplEx (KGE)	complex bilinear, dim 128 ($\rightarrow 256$)	triple	neg.-sampling rank	Adam	200	10^{-2}	32,768	66 m
HGT	2 layers, 4 heads, hidden 256, out 128	384/256	logit-adj. listwise CE	Adam (cos., wd 10^{-4})	300	10^{-3}	per-user	A100
KNN-CF	user-user cosine, $k=6$	interactions	—	—	—	—	—	< 1 m
XGBoost Hybrid	GBDT, 300 rounds (ES 30)	300	rank:ndcg@10	—	300rd	—	—	23 s

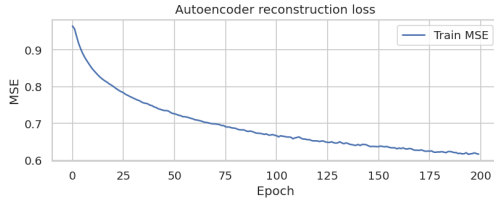


Figure 2. Autoencoder reconstruction MSE over 200 epochs (0.964 $\rightarrow$ 0.616).

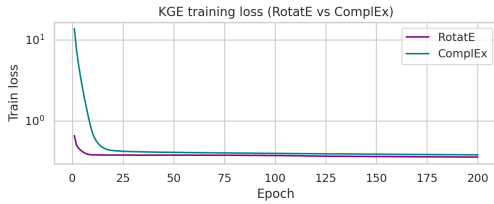


Figure 3. KGE training loss (log scale) for RotatE and ComplEx over 200 epochs. RotatE reaches a lower final loss (0.356 vs. 0.381).

7. Results

7.1. Representation Learning Curves

The autoencoder trains for the full 200 epochs (early stopping never triggers), with MSE falling from 0.964 to 0.616, steepest in the first ~ 75 epochs (Fig. 2). The residual MSE reflects the aggressive $12\times$ dimensionality reduction. RotatE converges to a lower absolute loss than ComplEx (0.356 vs. 0.381; Fig. 3), consistent with its better fit to the asymmetric/compositional relations in the KG.

7.2. HGT Training, Selection, and Ablations

Figure 4 summarises the HGT. Training and validation listwise loss decrease smoothly with no divergence (panel a); validation Overall@10 and its components rise and plateau, with the best checkpoint at epoch 240 (Overall@10 0.367; panel b). The debiasing ablation (panel c) reveals a clean monotonic trade-off: increasing λ lowers NDCG/Recall but raises coverage and lowers popularity bias. Concretely (vali-

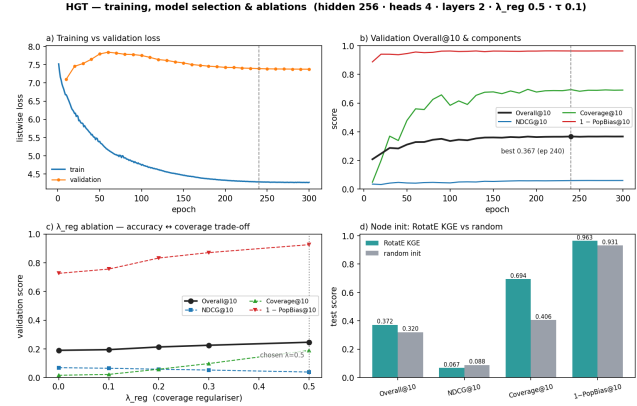


Figure 4. HGT (hidden 256, 4 heads, 2 layers, $\lambda=0.5$, $\tau=0.1$). (a) train/validation listwise loss; (b) validation Overall@10 and components, best at epoch 240; (c) λ accuracy-coverage trade-off; (d) RotatE vs. random node initialisation on test.

dation, selected configuration): $\lambda=0$ gives Recall@10 0.123, NDCG@10 0.069, coverage 1.6%, whereas $\lambda=0.5$ gives Recall@10 0.070, NDCG@10 0.038, coverage 18.9% and the best Overall@10 (0.246); checkpoint selection on the composite score therefore lands at $\lambda=0.5$. The initialisation ablation (panel d) shows that, *unlike our Milestone 2 finding*, RotatE now beats random initialisation on the composite objective (Overall@10 0.372 vs. 0.320), driven by markedly higher coverage (0.696 vs. 0.406); random init retains a small NDCG edge (0.088 vs. 0.067), confirming that KGE structure mainly helps diversity rather than raw ranking.

7.3. Benchmark

Table 3 reports all models at $K=10$ over the full 284,864-user test set, and Table 4 the multi- K sweep. **KNN-CF** attains the best Overall Score (0.421), combining competitive NDCG@10 (0.106) with 91.6% coverage. **The retrained HGT is a close second** (Overall 0.399): it *beats KNN-CF on per-user Recall@10* (0.105 vs. 0.103, $p < 10^{-5}$), reaches 79.7% coverage, and has the *lowest popularity bias of all models* (0.053 vs. 0.130 for KNN-CF). MostPopular leads raw NDCG@10 (0.114) but recommends from only 0.6%

Table 3. Test benchmark at $K=10$ (all 284,864 test users). Coverage and Popularity Bias (PB) are catalogue-level. Overall weights NDCG:Cov:(1−PB) as 0.6:0.2:0.2. Best per column in **bold**.

Model	Rec.	NDCG	HR	MRR	Cov.	PB	Ovrl.
MostPopular	.129	.114	.200	.090	.006	.611	.148
KNN-CF	.103	.106	.169	.087	.916	.130	.421
XGBoost	.038	.044	.072	.035	.667	.118	.336
HGT	.105	.097	.175	.074	.797	.053	.399

Table 4. Multi- K evaluation. Overall = 0.6 NDCG+0.2 Cov+0.2(1−PB).

Metric	Model	K=5	K=10	K=20	K=50	K=100
Recall	HGT	.069	.105	.155	.241	.324
	KNN-CF	.079	.103	.115	.121	.129
	MostPopular	.088	.129	.196	.303	.399
	XGBoost	.028	.038	.050	.069	.085
NDCG	HGT	.079	.097	.113	.133	.148
	KNN-CF	.095	.106	.109	.111	.113
	MostPopular	.096	.114	.135	.159	.177
	XGBoost	.038	.044	.049	.055	.059
Cov.	HGT	.716	.797	.871	.949	.984
	KNN-CF	.851	.916	.950	.959	.960
	MostPopular	.005	.006	.009	.014	.024
	XGBoost	.570	.667	.751	.826	.859
Overall	HGT	.372	.399	.425	.456	.474
	KNN-CF	.392	.421	.440	.451	.455
	MostPopular	.110	.148	.182	.232	.267
	XGBoost	.304	.336	.364	.389	.401

of the catalogue (Overall 0.148). XGBoost reaches 66.7% coverage but a retrieval bottleneck caps NDCG@10 at 0.044. All pairwise Recall/NDCG differences are statistically significant ($p < 0.001$ except HGT vs. KNN-CF on Recall, $p \approx 9 \times 10^{-6}$); Figure 5 shows per-user distributions with set-level significance bars. The multi- K sweep (Tab. 4) confirms the ordering: HGT’s Overall Score climbs from 0.372 at $K=5$ to 0.474 at $K=100$ and overtakes KNN-CF for $K \geq 20$ as its coverage and low popularity bias compound.

7.4. Latent Space and Interpretability

Projecting all nine node types from the HGT’s 128-dimensional output space to 2-D with UMAP (Fig. 6) reveals a strongly *type-structured* geometry: in most cases the model places each node type in its own region. Artists and the attribute/concept nodes — genres, keys, modes, tempo classes, instruments, and decades — collapse into a single, well-separated cluster (GMM cluster 13), distinct from the user–track mass. Within that mass, however, users and tracks are deliberately *not* pulled apart: a dense track region sits alongside the user clusters and individual users fall right among tracks. We read this user–track proximity not as

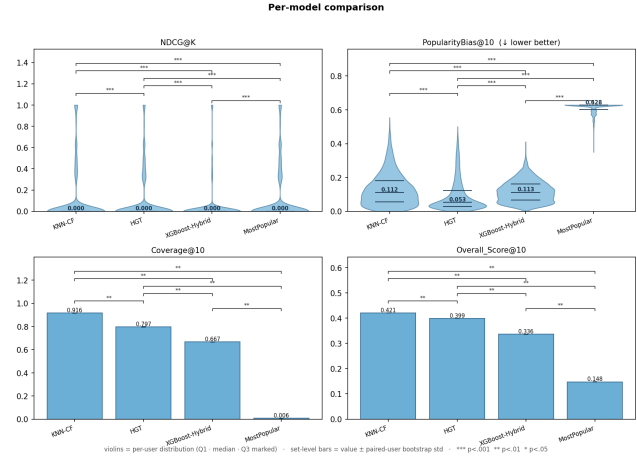


Figure 5. Per-user metric distributions with set-level significance bars ($***p<.001$). HGT has the lowest popularity bias (0.053) and the second-highest Overall Score (0.399), behind KNN-CF (0.421).

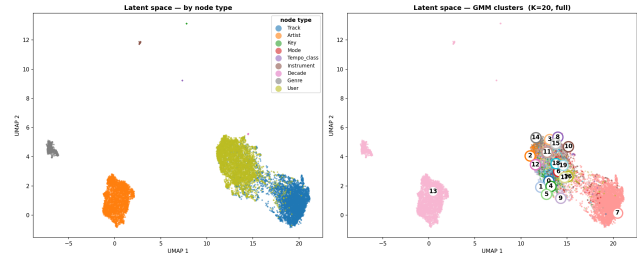


Figure 6. All-node-type HGT latent space (UMAP). Left: by node type; right: 20 BIC-selected GMM components. Artists and attribute/concept nodes form a well-separated region (cluster 13), while users and tracks share a common region and intermix — consistent with the cosine user–track scoring objective.

noise but as the geometric signature of the training objective — recommendation scores are cosine similarities between user and track embeddings, so the model is rewarded for embedding each user into the track manifold near the items it would recommend. In other words, on top of the clean content/structure separation, the HGT learns a collaborative-filtering-like placement of users among the tracks they are predicted to like.

Restricting the projection to users alone and fitting a BIC-optimal Gaussian Mixture yields 20 components (Fig. 7) that form coherent *listener archetypes* described by their dominant genre, tempo class, mode, and decade (e.g. “rock · Allegro · 2000s”, “dance pop · Andante · 2000s”, “rock · Allegro · 1970s”). That such structure emerges at all is evidence that HGT message passing propagates musically meaningful context rather than merely fitting interactions; these archetypes also seed the cold-start persona recommender of Sec. 7.5.

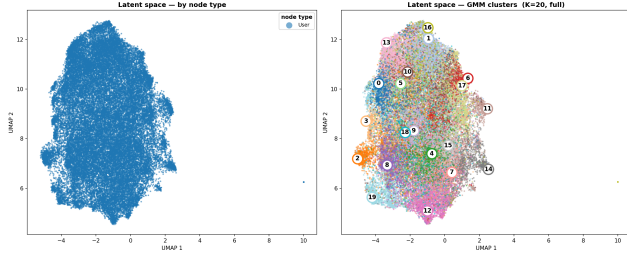


Figure 7. HGT user latent space (UMAP). Left: all users; right: 20 BIC-selected GMM components, each an interpretable listening archetype.

Caveats on the latent analysis. Two limitations temper this reading. First, the embeddings inhabit a 128-dimensional space whose spatial arrangement is not directly comprehensible to humans; the 2-D UMAP plots are a non-isometric, a projection that distorts true distances and densities, so they are a qualitative aid rather than a faithful picture of the latent geometry (the GMM is fit in the full 128-D space — UMAP is used only for display). Second, for the selected full-covariance model the BIC decreased monotonically up to the largest component count we tested (20) without reaching a minimum (Fig. 13, Appendix), so the 20-component choice is a boundary selection rather than a true optimum. We capped the sweep at 20 deliberately: a cold-start persona system that must slot a new user into one of *more* than twenty profiles defeats its own purpose, and our aim here was chiefly to confirm that the continuous embeddings cluster meaningfully in high dimensions rather than to identify the globally optimal mixture. Extending the component sweep or adopting a different clustering method (e.g. agglomerative/hierarchical clustering) is left to future work.

Because HGT learns typed attention, we can attribute each recommendation to specific neighbours and edge types. Figure 8 contrasts three attribution views — raw attention, score-delta ablation, and integrated gradients — for one recommendation, showing that shared instruments and a matching mode/genre drive the suggested track. Such rationales are produced at no extra modelling cost and are unavailable to the CF baselines.

At the population level (Fig. 14, Appendix), HGT attains by far the highest user–recommendation cosine alignment in its own latent space (mean 0.004 vs. ≤ 0.001 for MostPopular) while distributing hits across the hit-rate-vs-cosine plane more evenly than the popularity baseline. A per-user contrastive study (Tab. 8, Appendix) further shows cases where the HGT is the *only* model to score hits (e.g. user 15906: 4 hits vs. 0 for all baselines), supporting the view that the graph model surfaces content-consistent items the behavioural baselines miss. For the XGBoost hybrid, gain- and SHAP-based attribution (Figs. 15–16, Appendix)

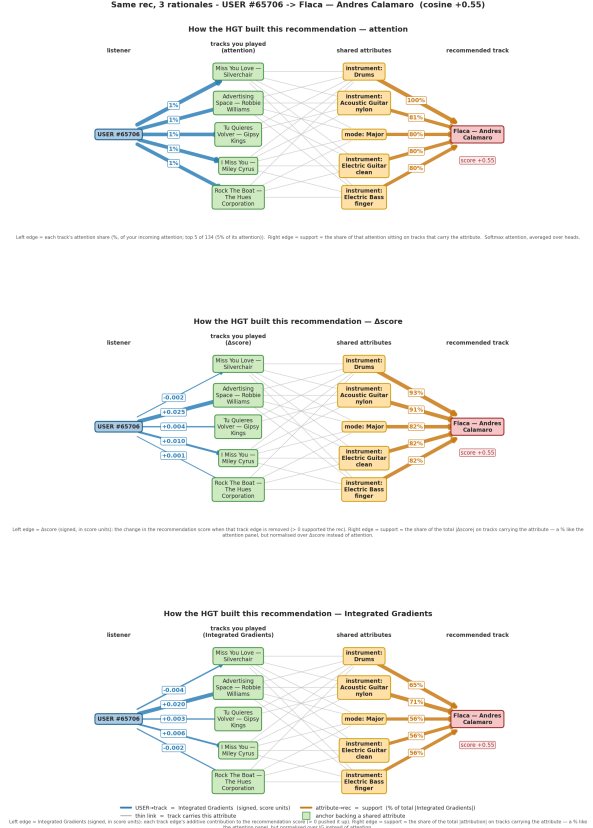


Figure 8. Three explanation views of a single HGT recommendation (attention, score-delta, integrated gradients), tracing the suggestion to shared instruments and a matching genre/mode.

show that the track autoencoder dimensions dominate ranking (mean|SHAP| 0.032 vs. 0.016 for the user profile and 0.007 for categorical features), with $\text{key}=\text{G}$ the single most influential categorical feature — confirming that content embeddings, not metadata, carry the hybrid’s signal.

7.5. Cold-Start Persona Recommendation

Every model evaluated above needs interaction history, so none can serve a brand-new user — the cold-start problem. We exploit a property unique to the HGT: because attribute nodes (genres, tempo classes, modes, decades, instruments), tracks, and users all share *one* embedding space, a new user can be placed in that space from a few declared preferences, with no retraining. The user selects musical attributes; we average the chosen anchor embeddings into a taste vector and offer three inference paths. (i) **Persona snap**: assign the vector to the nearest of 20 *listener archetypes* — the centroids of a users-only GMM fit on the HGT latent space (Sec. 7; examples in Tab. 7, Appendix). (ii) **k-NN user blend**: form a bespoke point on the user manifold as a similarity-weighted softmax mean of the $k=15$ nearest real users, with neighbour dispersion as a confidence signal. (iii) **Inductive HGT**:

Table 5. HGT under directed vs. consistent undirected inference (test set). The directed/undirected mismatch alone, not model capacity, explains the Milestone 2 collapse.

Metric	Undirected	Directed	Δ
Recall@10	0.105	0.001	0.104
NDCG@10	0.097	0.002	0.095
HitRate@10	0.175	0.003	0.172
Coverage@10	0.796	0.041	0.755
Popularity Bias	0.090	0.009	0.082
Overall Score	0.399	0.208	0.192

insert a temporary user node into the undirected graph, feed it a KGE-space feature from the picked attributes, wire it to a few taste-matched seed tracks, and run a single forward pass; this yields an inductive user embedding *and* genuine softmax attention, so the same explainer used in Sec. 7 produces faithful per-edge rationales for the new user. All three are pure inference and are exposed through an interactive widget (taste form $\rightarrow$ ranked recommendations $\rightarrow$ “why” panel), turning the learned latent geometry into a usable cold-start recommender.

8. Discussion: Known Issues and Limitations

Resolving the Milestone 2 evaluation artefact. Our Milestone 2 report recorded a near-zero HGT benchmark caused by a train/inference mismatch: the model was trained with undirected message passing but evaluated on the directed graph, which prevents messages from reaching user nodes at inference. Re-running inference on the undirected graph — matching training — recovers the full performance (Tab. 5): Overall Score rises from 0.208 to 0.399 and Recall@10 from 0.001 to 0.105. This single protocol fix, not any architectural change, is what moves the HGT from last to second place; all numbers in this report use the consistent undirected protocol.

Offline evaluation is an incomplete proxy. The composite score exposes a genuine ranking-versus-diversity tension: MostPopular maximises NDCG by exploiting long-tail play counts yet ignores 99.4% of the catalogue, while HGT and KNN-CF trade a little NDCG for near-complete coverage. Crucially, a track’s absence from a user’s history does not imply dislike, so a held-out implicit-feedback test set undercredits models (like HGT) that surface unheard but plausibly-liked items — consistent with HGT’s high latent alignment despite a modest hit rate, and motivating a user study.

Retrieval bottleneck. The XGBoost hybrid’s low NDCG is a first-stage ceiling from 200-candidate retrieval, not a re-ranking failure; an approximate nearest-neighbour candidate stage is the highest-value fix.

KGE contributes diversity, not ranking. RotatE initialisation now improves the composite score via coverage, but random init still wins on raw NDCG, so the KGE structural prior is partly subsumed by HGT’s own message passing. A controlled study of frozen vs. fine-tuned KGE and of corpus composition is warranted. Importantly, only RotatE was ever plugged into the HGT — ComplEx was trained but never used as an initialiser — so a natural next step is to compare a *spectrum* of KGE algorithms as initialisers and observe how each inductive bias propagates through message passing.

Limited search. Hyperparameters were chosen by a coarse one-axis sweep under time pressure; a joint Bayesian search over width, depth, heads, τ , and λ could shift the operating point.

9. Conclusion and Future Work

We built a deep pipeline — symbolic autoencoder, transductive KG embeddings, and a heterogeneous graph transformer with a popularity-debiased listwise loss — for content- and preference-aware music recommendation. Evaluating the HGT with message passing consistent with training (undirected) fixes the Milestone 2 evaluation artefact, and a re-tuned architecture lifts it further: the model is now a strong second overall (0.399), beats KNN-CF on Recall@10, overtakes it on Overall Score for $K \geq 20$, and achieves the lowest popularity bias of any model while remaining interpretable. Future work includes ANN-based candidate generation for the hybrid, a factorial ablation isolating KG structure / symbolic features / debiasing, graph-collaborative baselines (LightGCN [12], KGCN [10]), initialising the HGT with alternative KG embeddings that span the inductive-bias spectrum — ComplEx (complex-valued, asymmetry by conjugation rather than rotation) and models that do or do not capture edge asymmetry and node-type distinctions, from translational (TransE [14]) and bilinear (DistMult [15], symmetric by construction) to tensor-factorisation (RESICAL [16]) — joint hyperparameter optimisation, a larger-component or hierarchical clustering of the latent space, and an online user study to address the offline-evaluation gap.

10. Member Contributions

Pedro Fanica (54346) Pipeline architecture and environment bootstrap, GraphDB upload and SPARQL validation, KG $\rightarrow$ PyG conversion, HGT model and training loop with the logit-adjusted listwise loss, architecture/ λ /init ablations, latent-space GMM/UMAP analysis and cold-start persona recommender, attention/IG explanation rationales.

Aarthi Perumpillichira (66404) KG design and construction: OWL ontology schema, `kg_builder`, Wikidata

entity alignment, decade temporal chain, rich-KG reification, SPARQL query library and GraphDB inspection.

Maja Skwaroń (66083) Data pipeline: LMD–MSD linking, Taste Profile filtering, jSymbolic/music21 feature extraction, the symbolic-feature autoencoder, KNN-CF and XGBoost baselines, the evaluation protocol and statistical significance testing.

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Appendix: Additional Results

A. Data and Knowledge-Graph Statistics

Table 6. Knowledge-graph composition: instance counts per node type (left) and per relation (right). The graph totals 1.91×10^7 triples over 304,110 entities and 29 relation types.

Node type	#	Relation	#
Listener	285,037	rdf:type	612,827
Track/Perf./Event	7,013	instrument	56,891
Artist	3,668	factor	56,891
Genre	645	hasGenre	18,136
Instrument	129	subClassOf	12,167
Key	12	performer	7,013
Tempo class	9	hasKey/Mode	7,013
Decade	7	hasTempoClass	7,013
Mode	2	inDecade	4,768

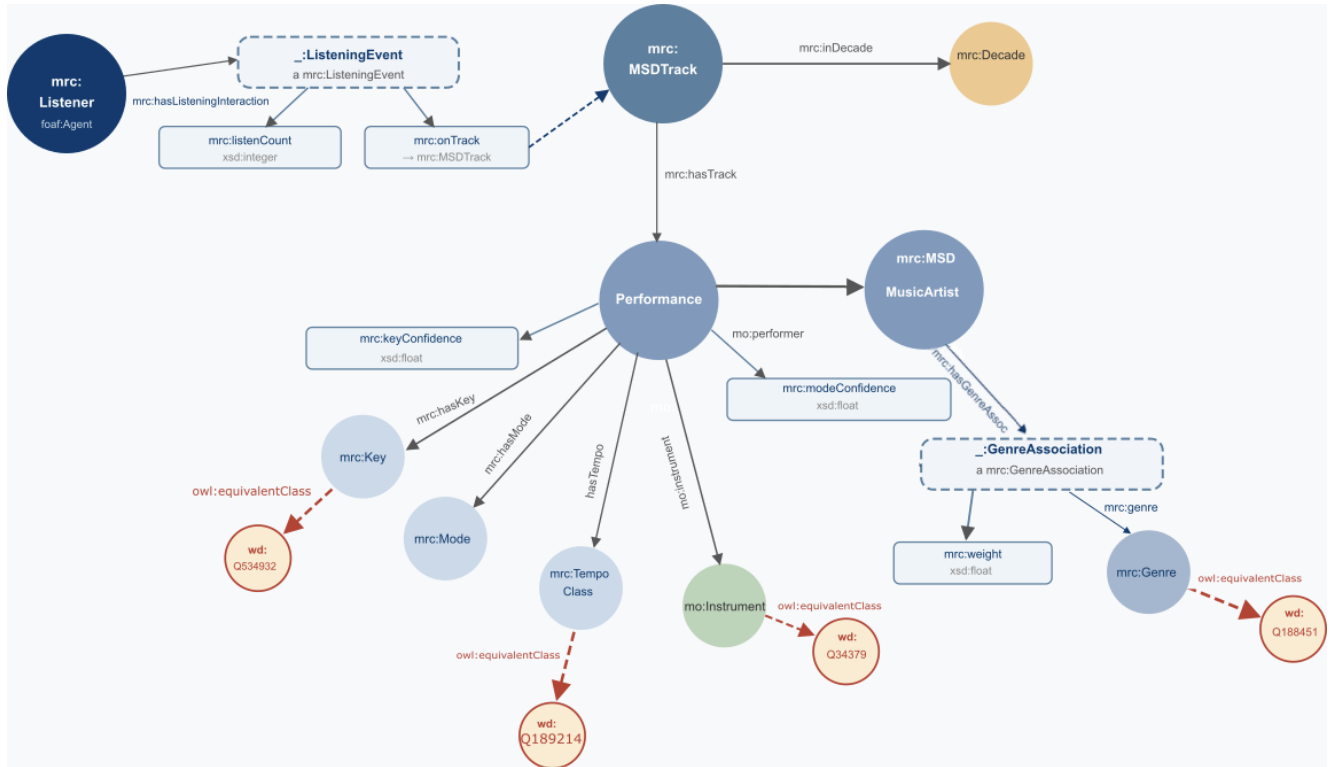


Figure 9. Schema-level ontology (TBox). `mrc:Listener` (a `foaf:Agent`) participates in `ListeningEvents` on `MSDTracks`; each track has a `Performance` carrying key, mode, tempo class, instruments, and a reified `GenreAssociation` with a weight. Structural classes are aligned to Wikidata via `owl:equivalentClass` (e.g. `Key` $\equiv$ `wd:Q534932`, `Instrument` $\equiv$ `wd:Q34379`, `Genre` $\equiv$ `wd:Q188451`).

B. Model Curves and Explanations

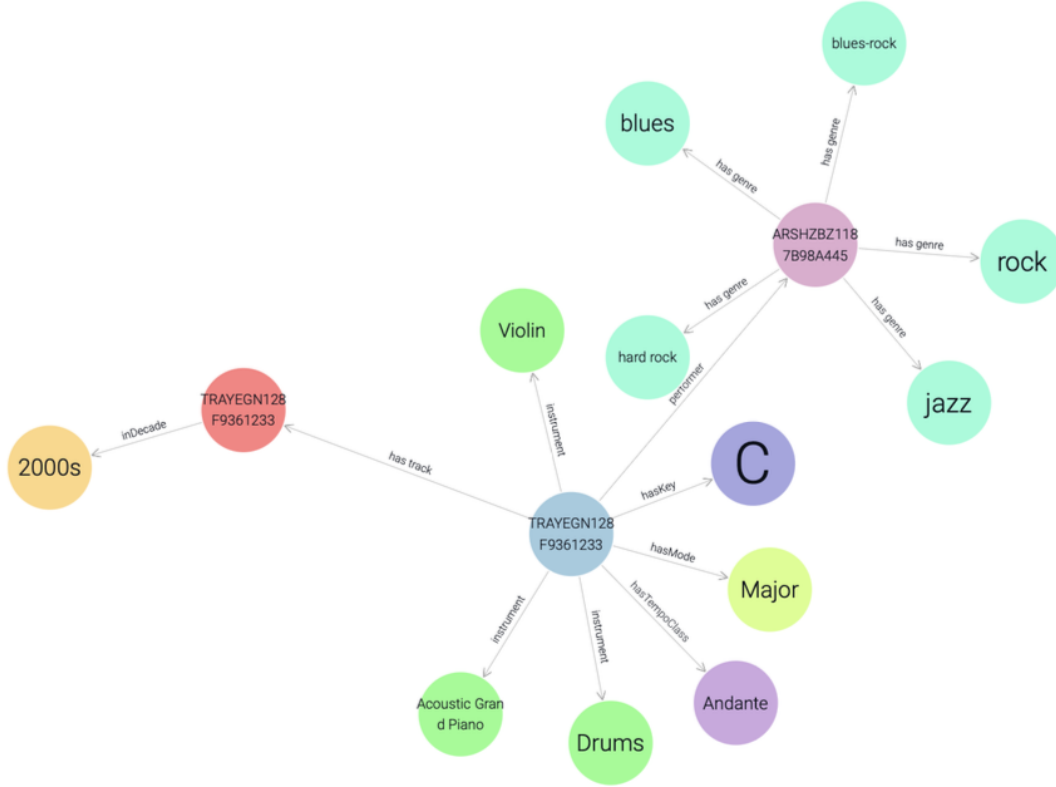


Figure 10. Concrete instance neighbourhood (ABox): a single track linked to its decade (2000s), key (C), mode (Major), tempo class (Andante), three instruments, and its artist, who in turn connects to genres (blues, blues-rock, hard rock, rock, jazz).

Table 7. Example HGT listener archetypes (users-only GMM centroids, labelled by their nearest tracks’ dominant genre, tempo class, and decade) used for cold-start persona snapping. Twenty archetypes are learned in total.

ID	Archetype (genre · tempo · decade)	# users
1	rock · Allegro · 2000s	18,244
5	rock · Andante · 2000s (hip hop/rap)	13,375
12	rock · Allegro · 1970s	19,238
17	dance pop · Andante · 2000s	12,791
19	rock · Allegro · 2000s (metal/hard rock)	13,430

Table 8. Per-user contrastive case studies (hits@10 per model; all “heavy” profiles). Selected to expose disagreement between models. The HGT is sometimes the only model with any hits (user 15906), illustrating complementarity with the behavioural baselines.

Case	n_{tr}	MostPop	KNN	XGB	HGT
hgt-win-heavy (u15906)	24	0	0	0	4
hgt-edge (u123955)	62	0	1	1	4
baseline-win (u204163)	79	5	8	0	0
all-success (u19316)	102	6	7	1	2
all-fail (u155906)	81	0	0	0	0

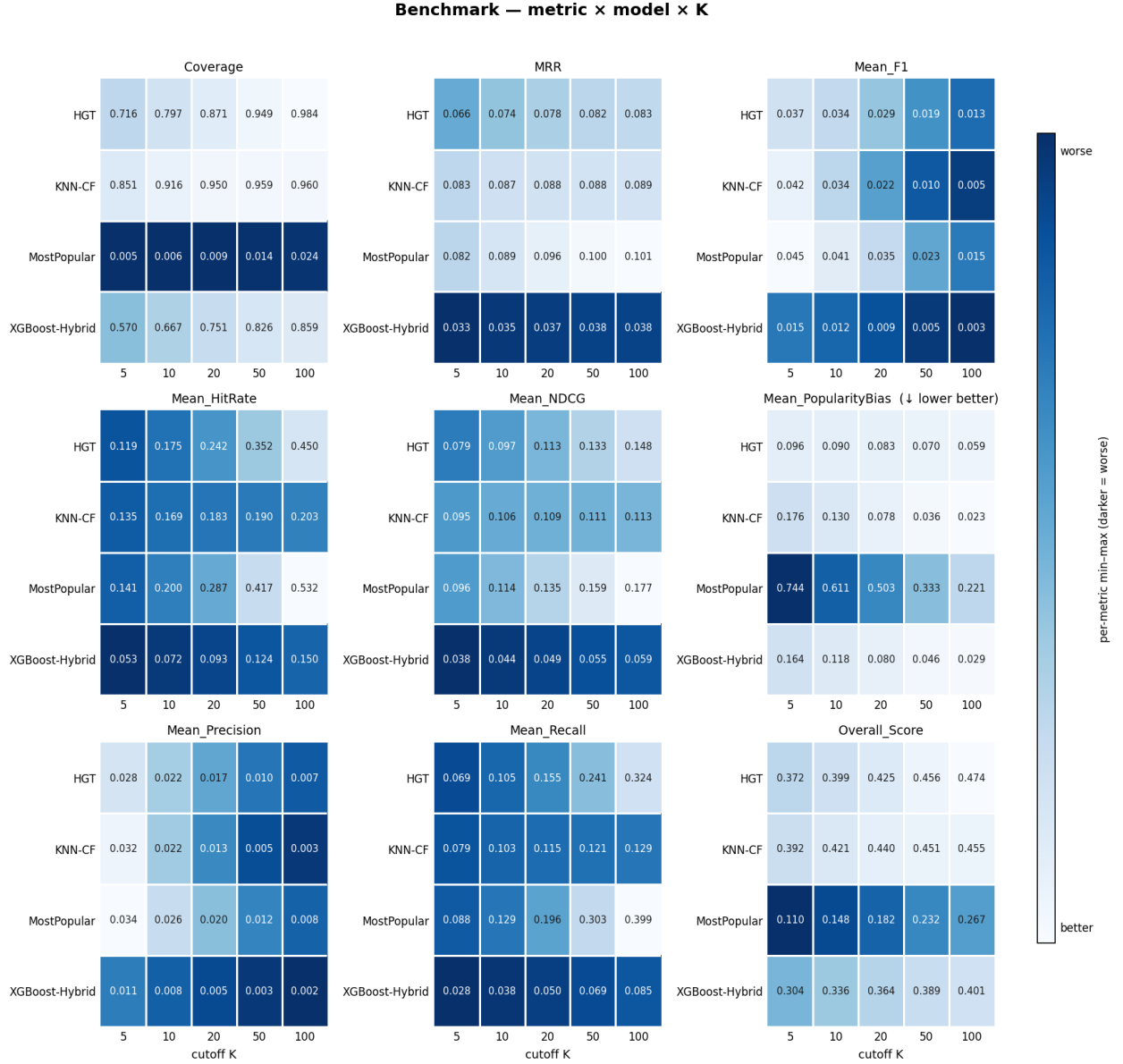


Figure 11. Full metric × model × K benchmark heatmaps. Darker is better per cell (popularity bias inverted). The HGT row is competitive across all K, dominating coverage and popularity bias among the high-NDCG models.

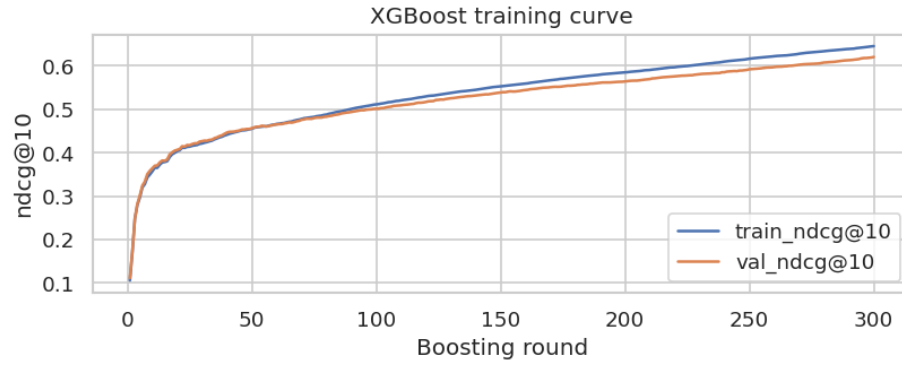


Figure 12. XGBoost LTR training curve (rank:ndcg@10, up to 300 rounds, early stopping 30). Rapid convergence confirms the 200-candidate retrieval bottleneck, not under-training, limits NDCG.

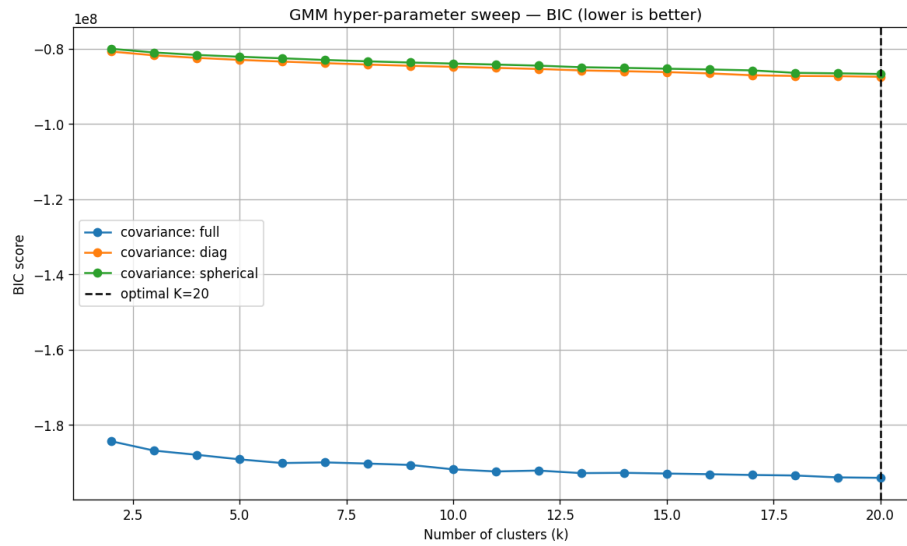


Figure 13. BIC over GMM component count for the HGT user latent space; the minimum selects 20 components.

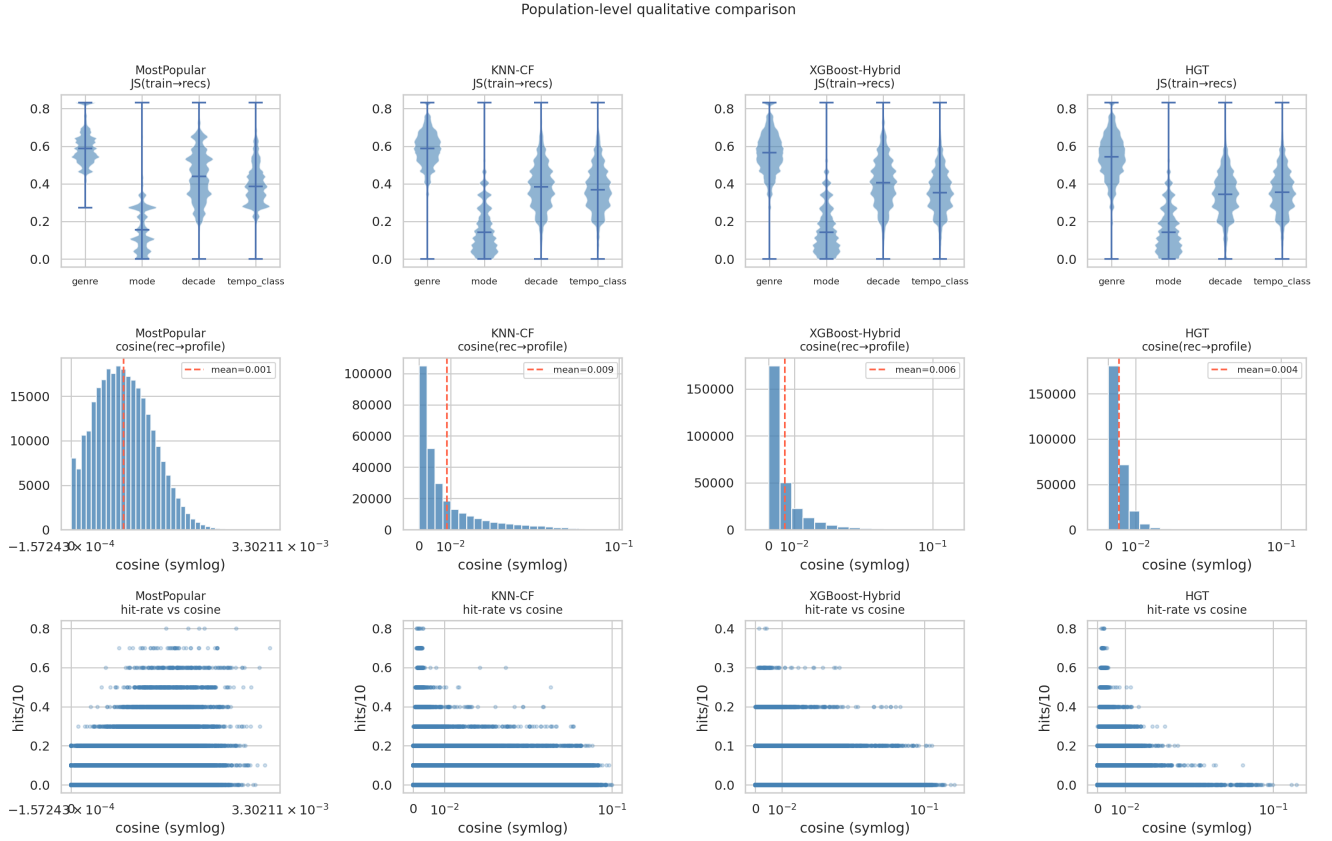


Figure 14. Population-level qualitative comparison across all four models. Rows: (top) attribute-distribution violins of recommended genre/mode/decade/tempo vs. the training profile; (middle) distribution of user–recommendation cosine similarity in the HGT latent space (HGT mean 0.004 vs. ≤ 0.001 for MostPopular); (bottom) hit-rate vs. cosine. HGT concentrates probability mass at higher cosine alignment than the behavioural baselines.

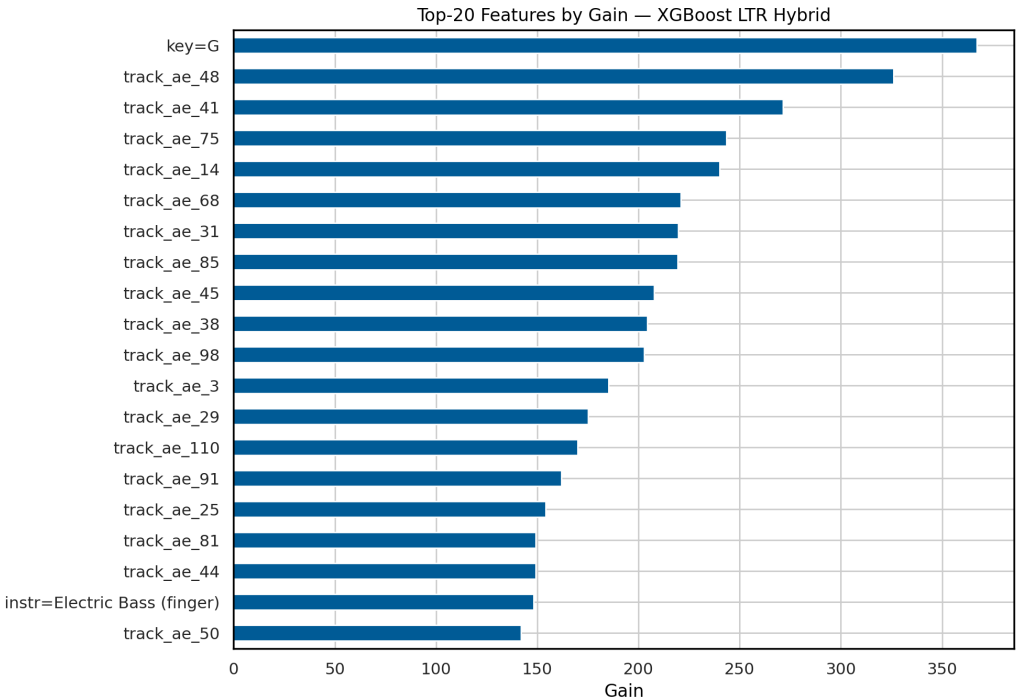


Figure 15. XGBoost hybrid: top-20 features by gain. Track autoencoder dimensions dominate, with `key=G` the strongest categorical feature.

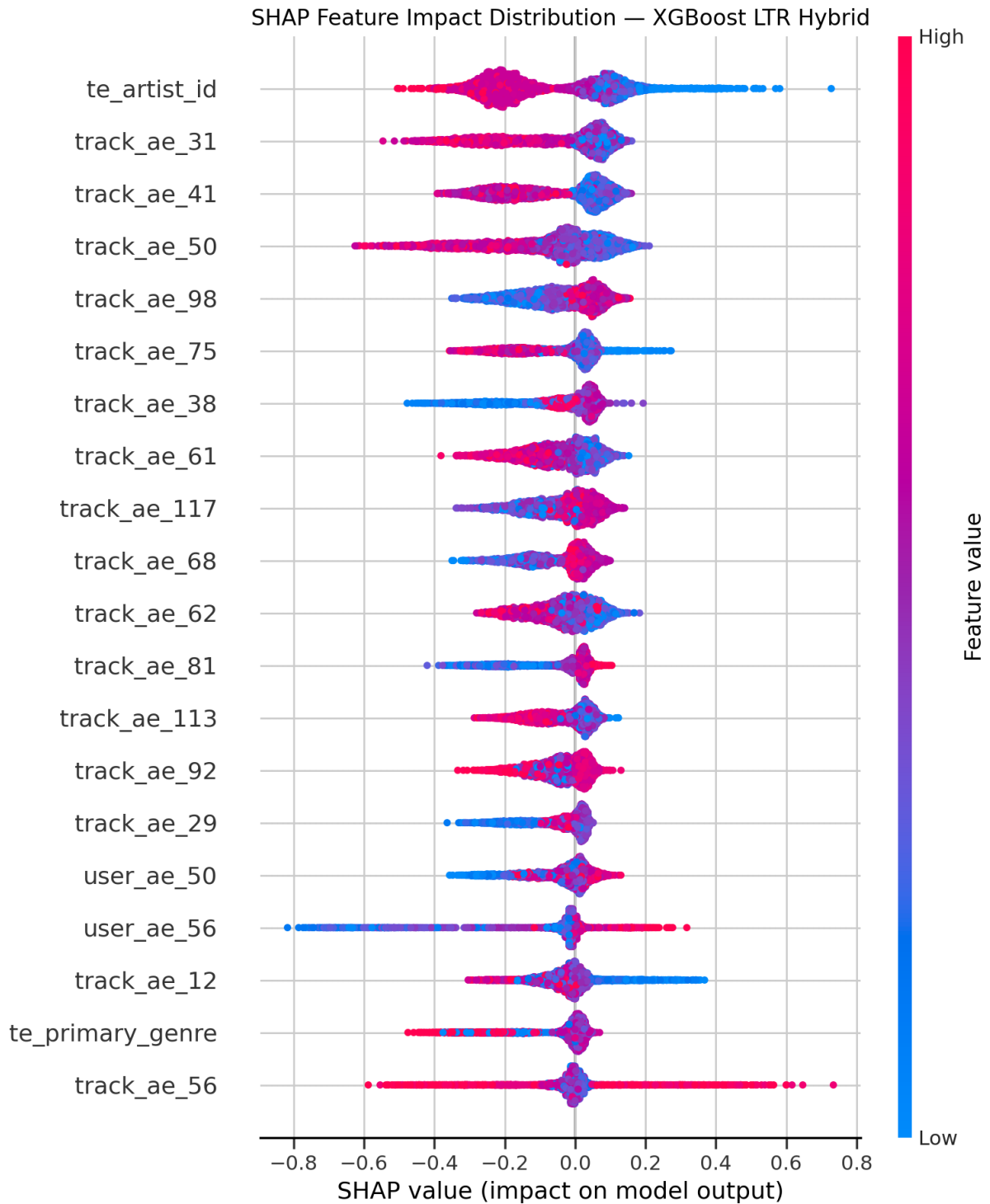


Figure 16. XGBoost hybrid: SHAP summary over a 1,000-user sample. Mean|SHAP| by block: track-AE 0.032, user-profile 0.016, categorical/content 0.007 — content embeddings drive the re-ranking far more than metadata.